MILITARY SPECIFICATION

AMPLIFIER-POWER SUPPLY AM-4080/ASN-55, LEVELING

1. SCOPE

1.1 This specification covers one type of leveling amplifier - power supply, designated AM-4080/ASN-55.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-D-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-M-26512	Maintainability Program Requirements for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-R-27542	Reliability Program for Systems, Subsystems and Equipment
MIL-A-38329	Attitude Heading Reference Set AN/ASN-55
MIL-G-38330	Gyroscope, Displacement CN-990/ASN-55
MIL-G-38332	Gyroscope, Rate Switching CN-1050/ASN-55
MIL-D-70327	Drawings, Engineering and Associated Lists

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STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MS3114	Connectors, Receptacle, Electric, Solder Type, Rear Mounting with Nut, Bayonet Coupling
MS17322	Meter, Time Totalizing Miniature, 400 Cycle, Digital
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The amplifier-power supply shall consist of a transformer to convert 115V a-c 3-phase 400-cps wye power to 115V a-c 400-cps 3-phase delta power; a servo amplifier and a transformer for controlling the system leveling axis; a thermal time delay relay to control automatic synchronizing functions of the system and a relay to provide an external fast erection function; and associated switching.

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-Proof Materials.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent. If used within a hermetically sealed enclosure, fungicidal treatment shall not be required.

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3.4.2 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.4.2.1 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.3 Nonmagnetic Materials.- Nonmagnetic materials shall be used for all parts of the amplifier-power supply except where magnetic materials are essential.

3.4.4 Corrosive Fumes.- Materials, as installed and used in the amplifier-power supply under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.5 Protective Treatment.- When materials used in construction of the amplifier-power supply are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction.- The amplifier-power supply is a component of the AN/ASN-55 attitude heading set conforming to MIL-A-38329. It shall:

a. Provide conversion from 115V ac 400-cps 3-phase wye to 115v ac 400-cps 3-phase delta

b. Provide fixed and variable voltage outputs for an external torque motor load

c. Provide an automatically timed fast erection initiating voltage for an external MD-1 gyroscope

d. Provide relay circuitry for control of certain functions of the using system in response to commands from the system.

3.5.1 Adjustments and repairs.- The amplifier-power supply shall be so designed and constructed that adjustments and repairs can be easily made by the personnel of overhaul bases.

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3.5.2 The amplifier-power supply shall be so constructed as to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage installation, and service.

3.5.3 Reliability.- The amplifier-power supply shall be so designed that when tested as specified in 4.6.20, the mean-time-between-failures (MTBF) will be at least 7,000 hours. The test program shall be based on a discrimination ratio of 0.33 and a confidence level of 90 percent. MIL-R-27542 shall be used as a guide for proper implementation of the required reliability into the design and construction of the amplifier-power supply.

3.5.3.1 Longevity.- The amplifier shall have an operating life span (equipment longevity as defined in MIL-R-26667) of not less than 2,000 hours before the equipment consistently fails to meet the MTBF index.

3.5.4 Maintainability.- The following maintainability requirements shall be considered in the design of the amplifier-power supply.

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspections, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly, or component

d. Minimum numbers and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal, or replacement.

3.5.5 Compatibility.- The amplifier-power supply shall be compatible with other equipments listed in MIL-A-38329.

3.5.6 Electronic components.- Except as specified herein, electronic components and the application thereof shall be in accordance with MIL-E-5400 for class 2 equipment, except that it shall be suitable for altitudes up to 80,000 feet.

3.5.7 Electrical power.- The amplifier-power supply shall be designed to operate from 27.5V dc and 115V 400-cps 3-phase a-c power in accordance with MIL-STD-704, category B.

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3.5.8 Signal input.- The servo amplifier in the unit shall be designed to operate from signals supplied by the synchro transmitter in the CN-990/ASN-55 displacement gyroscope conforming to MIL-G-38330.

3.5.9 Outputs.- The amplifier-power supply shall provide the following outputs:

a. 3-phase 400-cps 115V delta power at up to 32 va per phase

b. A 27V 400-cps output to the fixed phase of the leveling torquer in the CN-990/ASN-55 displacement gyroscope

c. A 400-cps output voltage to the control winding of the leveling torque motor in the CN-990/ASN-55 displacement gyroscope, controlled by the leveling synchro therein

d. A 60-second timed 115V 400-cps signal initiated by a ground pulse

e. A 115V 400-cps signal that responds to the CN-1050/ASN-55 rate switching gyroscope conforming to MIL-G-38332 to cut out roll erection of an external displacement vertical gyroscope.

3.6 Performance.- The amplifier-power supply shall be capable of operating as specified herein under the following conditions:

a. Temperatures: Operating (continuous or intermittent) temperatures ranging from -54° to +71°C and storage temperatures ranging from -62° to +85°C

b. Altitude: Up to 80,000 feet

c. Vibration: Up to 5g's, operating

d. Acoustical noise: Up to 140 db

e. Acceleration: Up to 10g's, operating

f. Relative humidity: Up to 100 percent

g. Dynamic shock: 10g's operating and 30g's nonoperating

h. Fungus: Exposure to fungus enzymes

i. Explosion: Operation in explosive atmosphere

j. Rain: Exposure to simulated rainfall of 4 inches per hour

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k. Salt fog: Exposure to salt fog for a period of 48 hours

l. Sand and dust: Exposure to blowing fine sand and dust particles.

3.6.1 Dielectric strength.- There shall be no insulation breakdown when 500V dc is applied between isolated circuits and the case or connector shell for a period of 10 seconds. The resistance shall be at least 50 megohms.

3.6.2 Radio Noise interference.- The amplifier-power supply shall meet the radio noise interference requirements of MIL-I-6181.

3.6.3 Warmup.- The period required for warmup shall not exceed 90 seconds.

3.7 Part numbering of interchangeable parts.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.8 Time totalizing meter.- An MS17322-8,10,000-hour time totalizing meter in accordance with MIL-M-7793 shall be provided in the amplifier-power supply.

3.9 Case.- The case shall be made of suitable material, shall be uniform in texture, and shall be entirely covered with a durable dull black finish. The case shall be so designed that the internal mechanism may be removed from the case and replaced without the use of special tools or fixtures unless they are approved by the procuring activity.

3.10 Electrical connector.- An MS3114P-18-32P (J200) electrical connector shall be provided on the unit. It shall be located as shown on figure 1 and shall be wired as follows:

Pin	Use
A	Phase A }
B	Phase B } 115/200V ac 400-cps 3-phase
C	Phase C } wye power input
D	Neutral
E	Spare
F	28V d-c input
G	Leveling pickoff (high), input
H	Leveling pickoff (low), input
J	115V ac (after start), input

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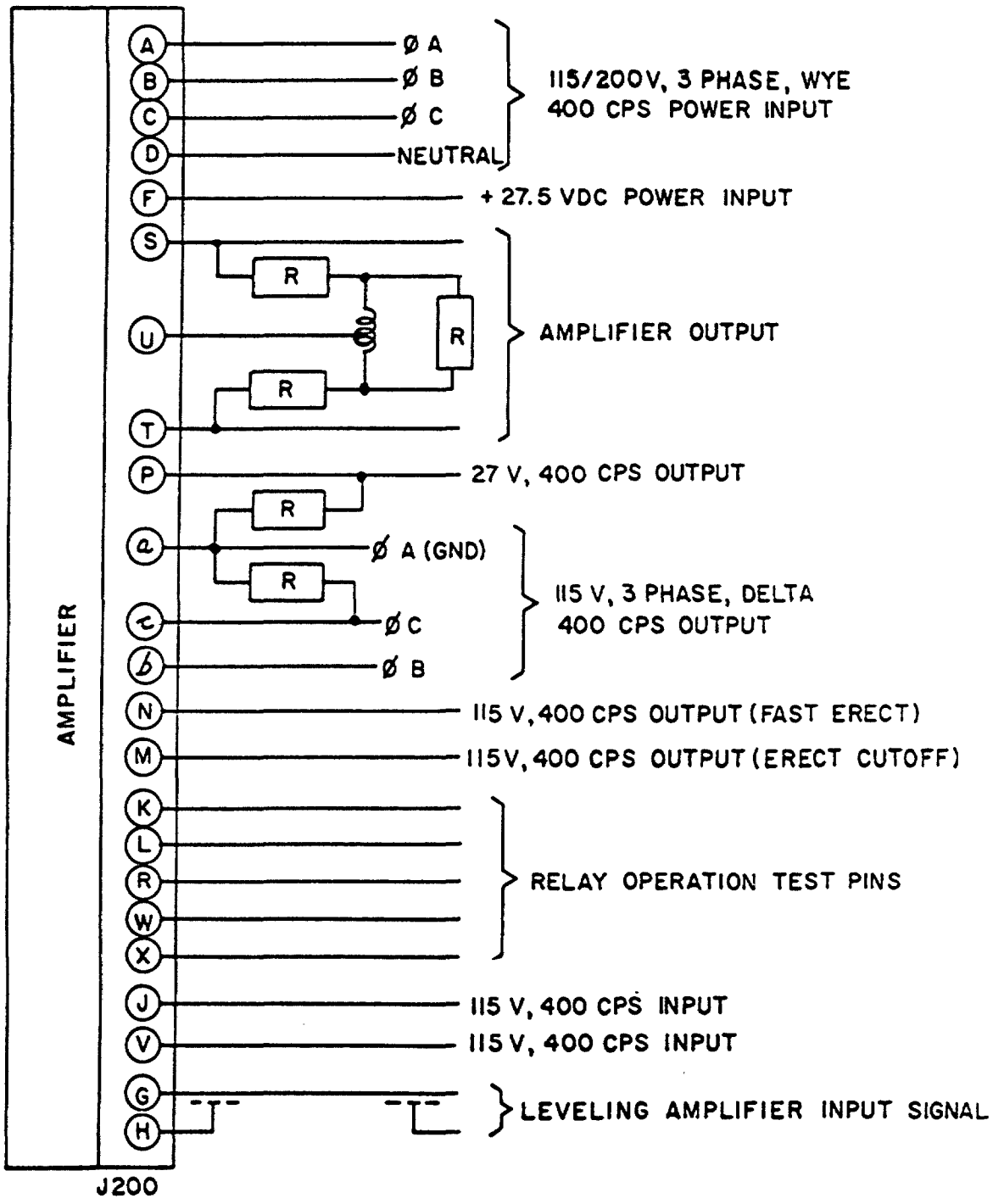


FIGURE 1. AMPLIFIER-POWER SUPPLY, AM-4080/ASN-55

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Pin	Use	
K	Erection switch, output	
L	Indicator flag, output	
M	Roll erection cutout, output	
N	Automatic fast erection, input	
P	Leveling torquer excitation output	
R	Indicator flag, output	
S	Leveling torquer	Output
T	Leveling torquer	
U	Leveling torquer CT	
V	Rate switching gyroscope, input	
W	Sync switch, output	
X	Automatic sync, output	
Y	Spare	
Z	Spare	
a	Phase A neutral	Power output 115V ac delta 3 phase 400-cps
b	Phase B	
c	Phase C	
d	Spare	
e	Spare	
f	Spare	
g	Spare	
h	Spare	
j	Spare	

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3.11 Electrical functions.- The amplifier-power supply shall provide the following functions when properly loaded and connected to electrical power and input signal sources.

3.11.1 Amplifier.- The amplifier shall provide the following outputs under normal conditions:

Input Voltage (RMS)	Output Voltage (RMS)	Phase
0	1.0 max	---
0.05	15.8 ±0.8	90° ±15° lead
0.08	18.0 min	---

3.11.2 Power supply.- The power supply shall provide the following outputs when properly loaded and supplied with input power:

a. 3-phase delta 400 cps 115V line-to-line with phases A, B, and C on pins a, b, and c of J200 with pin a grounded. Phase rotation shall be A, B, C

b. 27 ±3V rms 400 cps between pins P and a with pin a grounded.

3.11.3 Fast erection control voltage.- When pins W and a of J200 are momentarily connected, 115V ac 400 cps shall appear between pins N and a of J200 for a period of 50 to 70 seconds.

3.11.4 Roll erection cutout.- Upon application of a 115V 400-cps signal to pin J200-V, 115V 400 cps, phase P shall appear between pins J200-M and J200-a. During this condition, the voltage specified in 3.11.3 shall not appear.

3.11.5 Flag and synchronization circuits.- With normal power applied, initiating the delay period specified in 3.11.3 shall result in grounding pins J200-L and J200-R and opening the circuit between pins J200-W and J200-X. The same shall result when a 115V 400-cps signal is applied to pin J200-J, regardless of the delay period specified in 3.11.3.

3.11.6 Power consumption.- During starting, the power consumption shall not exceed 58 va ac nor 4w dc. During operation, the power consumption shall not exceed 50 va ac nor 4w dc.

3.12 Dimensions and mounting.- The overall dimensions and mounting of the amplifier-power supply shall be in accordance with figure 2.

3.13 Weight.- The weight of the amplifier-power supply shall not exceed 4.0 pounds.

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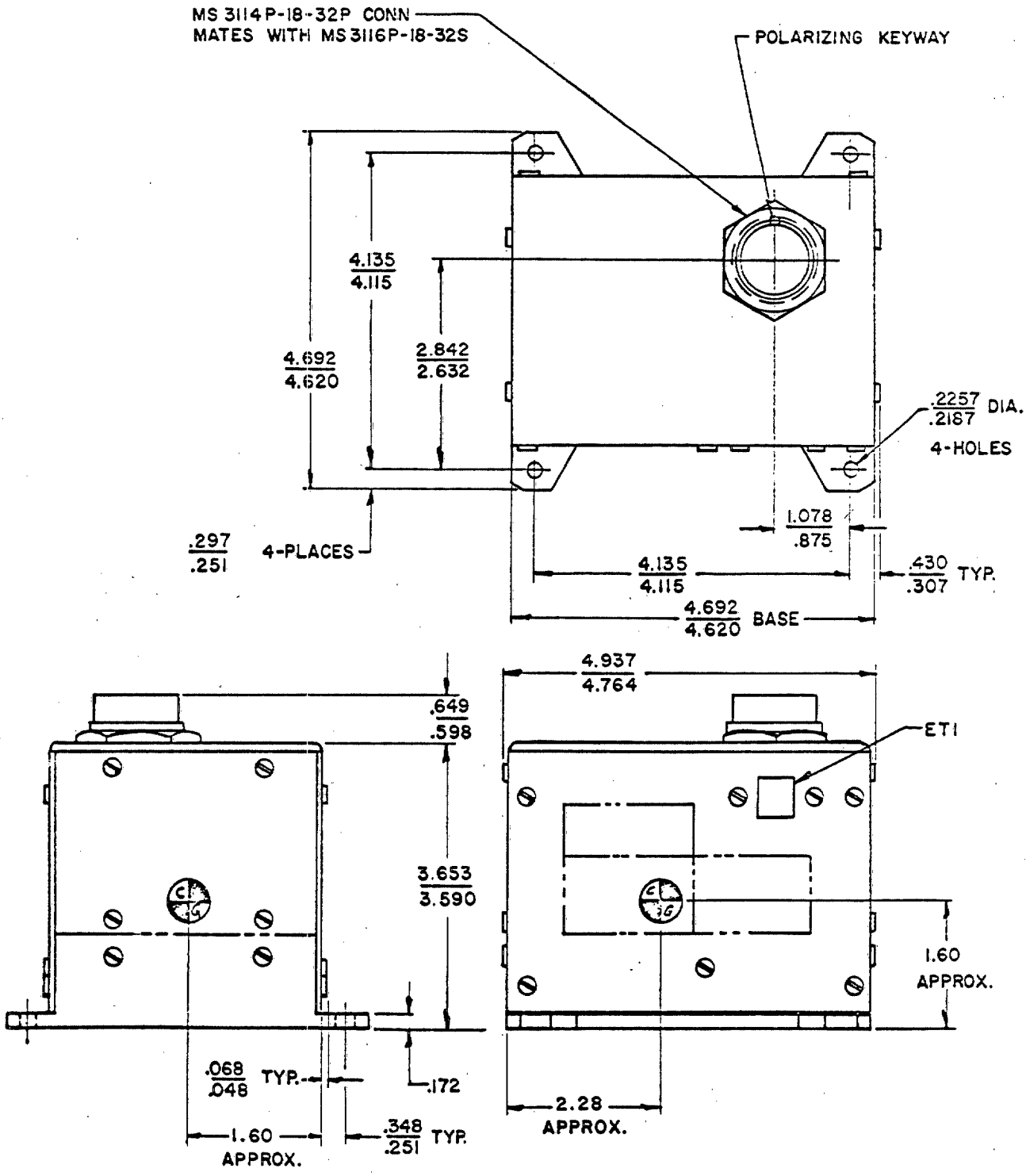


FIGURE 2. INSTALLATION DATA

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3.14 Soldering.- Soldering shall be accomplished in accordance with MIL-S-6872.

3.15 Screw Threads.- Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Finishes and Protective Coatings

3.16.1 Aluminum Alloy Parts.- Aluminum parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.16.1.1 Dials, small holes, and case inserts need not be anodized.

3.16.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.16.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.16.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.16.1.5 When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.16.2 Cadmium Plating.- Cadmium plating shall be in accordance with QQ-P-416 and of a type and class adequate to achieve the degree of protection required.

3.17 Identification of Product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.18 Workmanship.- The amplifier-power supply, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.18.1 Dimensions and Tolerances.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

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3.18.2 Screw Assemblies.- Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, or bolt, or threads.

3.18.3 Riveting.- Riveting operations shall be performed carefully to insure that the rivets are tight and satisfactorily headed.

3.18.4 Gears.- Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from backlash.

3.18.5 Cleaning.- The amplifier-power supply shall be cleaned thoroughly of loose, spattered, or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests.- The inspection and testing of amplifier-power supplies shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions.- Unless otherwise specified, the equipment shall be subjected to all tests under the following conditions:

4.3.1 Environmental.- The equipment shall be tested in the following environment:

a.	Temperature	25° ± 10°C
b.	Altitude	Normal ground
c.	Vibration	None

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d. Relative humidity	Up to 100 percent
e. Acceleration	At rest

4.3.2 Input power.- Tests shall be performed with the following input power. Application of the power shall be as specified under 4.6.

a. 3-phase 4-wire wye, 400 cps ±1 percent 200V ac rms ±2 percent line-to-line, 115V ac rms ±2 percent line-to-neutral with neutral grounded. Phase rotation shall be A, B, C and shall be connected to pins A, B, C, respectively, of J200 with neutral grounded to pin D of J200

b. 27.5V ±2 percent dc, positive connected to J200, pin F, and the negative to J200, pin D.

4.3.3 Input signals.- The following input signals shall be applied as specified under 4.6.

a. A signal of 115V ac rms ±2 percent 400 cps ±1 percent, phase B, connected between pins J and D of J200 with D neutral

b. A signal of 0.050V ac rms ±2 percent 400 cps ±1 percent in phase within ±3° with the voltage between pins C and D of J200 connected between pins G and H of J200 with H grounded. The source impedance of this signal shall be 100 ohms, maximum

c. A signal of 0.080V ac rms ±2 percent 400 cps ±1 percent in phase within ±3° with the voltage between pins C and D of J200 connected between pins G and H of J200 with H grounded. The source impedance of this signal shall be 100 ohms, maximum

d. A signal of 115V ac rms ±2 percent 400 cps ±1 percent (any phase) connected between pins v and D of J200 with D neutral.

4.3.4 Output loads.- The following output loads shall be connected as specified under 4.6.

a. A center tapped load having an end-to-end a-c impedance of 354 ohms ±2 percent +j274 ohms ±2 percent at 400 cps and a d-c resistance of 234 ohms ±10 percent connected between pins T and S of J200 with the center tap connected to pin U of J200. The impedance shall be maintained within the specified tolerance under all output

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levels resulting from the tests specified herein. A suggested dynamic range is 0.1 to 30.0V rms. The center tap position must be accurate within 2 percent for both impedance components. A satisfactory dummy load circuit is shown on figure 1

b. A resistive load of 450 ohms ±5 percent connected between pins c and a of J200. The load shall be capable of dissipating 35w

c. A resistive load of 500 ohms ±5 percent connected between pins P and a of J200. The load shall be capable of dissipating 3w.

4.4 Preproduction testing

4.4.1 Test samples.- The test samples shall consist of 18 amplifier-power supplies representative of the production equipment. Three of the units shall be subjected to all the tests except reliability and the other 15 units shall be subjected to the test specified in 4.6.20. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The 3 units shall be forwarded to the procuring activity after testing.

4.4.1.1 Reliability demonstration.- Unless otherwise specified, 15 amplifier-power supplies shall be selected at random from the first 50 produced on the contract or order and tested for a total of 23,100 hours unless rejection is determined in accordance with figure 3 prior to that time.

4.4.2 Test report.- A preproduction test report shall be prepared in accordance with MIL-STD-831 and two complete copies of the report furnished to the procuring activity.

4.4.3 Preproduction tests.- The preproduction tests shall consist of all the test specified under 4.6.

4.5 Acceptance tests.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

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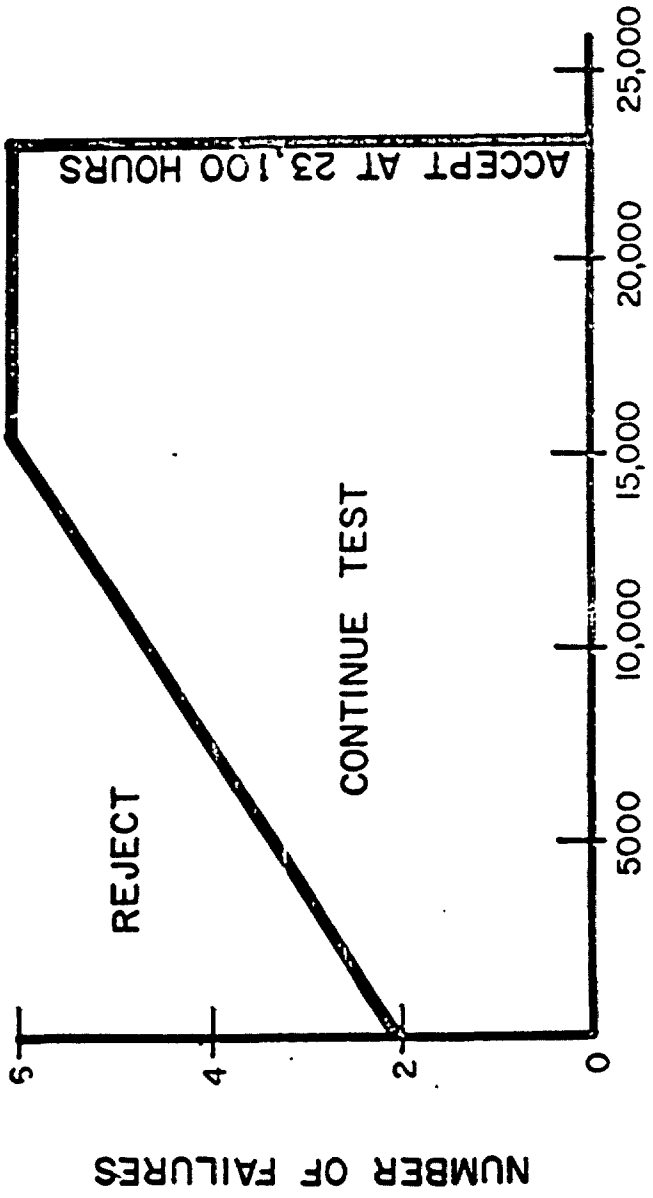


FIGURE 3. CONTINUE TEST - REJECT CRITERIA

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4.5.1 Individual tests.- Each amplifier-power supply shall be subjected to the following applicable tests as described under 4.6.

a. Examination of product

b. Continuity tests

c. Power supply

d. Relay operation

e. Leveling amplifier test

f. Time totalizing meter.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A.- One amplifier-power supply shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6.

a. Individual tests

b. Low temperature

c. High temperature

d. Vibration error

e. Voltage and frequency variation

f. Dielectric strength.

4.5.2.2 Sampling plan B.- Unless otherwise specified, 3 units shall be selected at random from the first 15 items on the contract or order and subjected to the following tests as described under 4.6.

a. Sampling plan A tests

b. Radio noise interference

c. Altitude-temperature

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d. Voltage - temperature

e. Acoustical noise

f. Dynamic shock

g. Vibration

h. Acceleration

i. Humidity

j. Salt fog

k. Sand and dust

l. Fungus

m. Explosion.

4.5.2.2.1 Rejection and retest.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all tests shall be repeated.

4.5.2.2.1.1 Individual tests may continue.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.5.3 Sampling plan C.- Unless otherwise specified, 13 amplifier-power supplies shall be selected at random from each 50 or less produced on the contract or order and tested for a total of 7,700 hours.

4.5.3.1 Acceptance or rejection.- Figure 4 shall be the criteria used to determine whether the equipment will be accepted or rejected. Certified time records of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.20 shall be maintained. In the event a continue-test decision applies after 7,700 hours of testing, the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to the data of the next production quantity for a composite accept-reject decision. If the continue test

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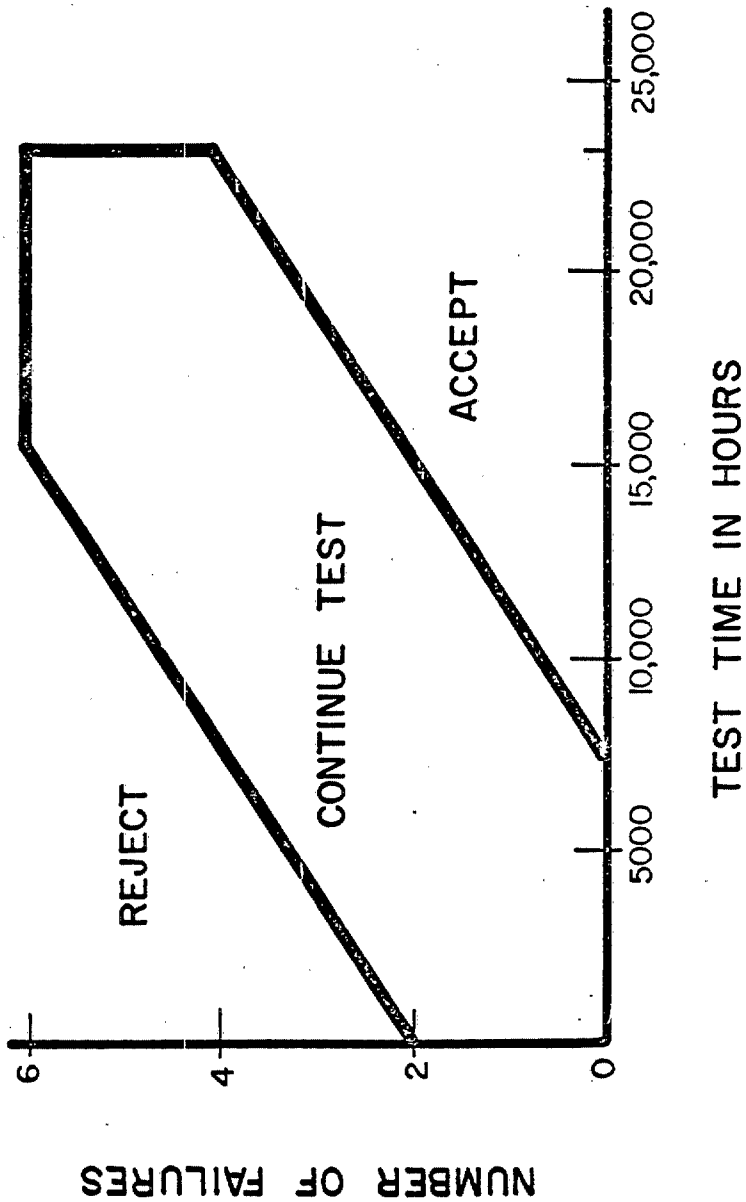


FIGURE 4. REJECT - CONTINUE TEST - ACCEPT CRITERIA

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condition persists, the data from successive production quantities shall be accumulated until 6 failures or 23,100 hours of testing have occurred at which time an accept-reject decision shall be made.

4.6 Test Methods

4.6.1 Examination of product.- The unit shall be inspected to determine compliance with figure 2, and material, workmanship, and marking.

4.6.2 Continuity tests.- With no power applied to the unit, the following continuities or discontinuities shall exist. Continuity shall be considered as a resistance of less than 1 ohm.

J200-B to J200-b	Continuity
J200-B to J200-J	Continuity
J200-F to J200-U	Continuity
J200-a to J200-D	Continuity
J200-W to J200-X	Continuity
J200-c to J200-N	Open
J200-B to J200-M	Open
J200-L to J200-D	Open
J200-R to J200-a	Open

4.6.3 Power supply.- With power applied per 4.3.2a and loads connected per 4.3.4a and 4.3.4b, output voltages shall be obtained as follows:

J200 pins	Output Voltage - rms	Phase
c to a	115 ±6	180° ±5° (Ref J200-A to J200-D)
P to a	27 ±3	0° ±5° (Ref J200-C to J200-D)

4.6.4 Relay operation.- Relay operation shall be observed with power applied per 4.3.2. Application of power shall not interrupt the continuity between pins J200-W and J200-X.

4.6.4.1 Pin J200-K shall be connected to pin J200-a momentarily. This test is not valid if repeated within 10 minutes.

4.6.4.1.1 For a minimum of 50 seconds after the connection of 4.6.4.1 is made, 115 ±5V rms, phase 180° ±5° (reference J200-A to J200-D) shall appear between pins J200-N and J200-a, with J200-a grounded, except as interrupted in 4.6.4.

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4.6.4.1.2 Within the 50-second period of 4.6.4.1.1, the following conditions shall exist:

J200-L to J200-a	Continuity
J200-R to J200-a	Continuity
J200-W to J200-X	Open

4.6.4.1.3 Within the 50-second period of 4.6.4.1.1, a signal per 4.3.3.d shall be applied. The voltage of 4.6.4.1.1 shall disappear and a voltage of 115 ±5V rms, phase 0° ±5° (reference J200-B to J200-D) shall appear between pins J200-M and J200-a, with pin J200-a grounded.

4.6.4.1.4 While still within the 50-second period of 4.6.4.1.1, the signal of 4.3.3.d shall be removed. The voltage of 4.6.4.1.3 shall disappear and the voltage of 4.6.4.1.1 shall reappear.

4.6.4.1.5 Not more than 70 seconds after the momentary connection of 4.6.4.1, the voltage of 4.6.4.1.1 shall disappear.

4.6.4.2 After applying a signal per 4.3.3.a, the following conditions shall exist. The signal of 4.3.3.1 and the power specified in 4.3.2 shall then be removed.

J200-L to J200-a	Continuity
J200-R to J200-a	Continuity
J200-W to J200-X	Open

4.6.5 Leveling amplifier.- After applying power per 4.3.2.b, signals per 4.3.3.b, then per 4.3.3.c, and the load per 4.3.4.a, the following output voltages shall be obtained.

Input Signal	Output Voltage RMS	Phase Reference (J200-C to J200-D)
0 (Pins J200-G and J200-H shorted)	1.0 maximum	---
Per 4.3.3.2	15.8 ±.8V	90° ±15° Lead
Per 4.3.3.3	18.0 minimum	---

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4.6.6 Time totalizing meter.- The time totalizing meter shall indicate the elapsed time that power is applied to the unit.

4.6.7 Low temperature.- Each sample unit shall be placed in a test chamber at a temperature of -54° ±2°C without power applied for a 4-hour period. At the end of this period and while at the low temperature, normal power (see 4.3.2) shall be applied. The unit shall meet the requirements of 4.6.3 through 4.6.6 within the following limits:

a. The delay time of 4.6.4.1 shall be from 45 to 85 seconds

b. Leveling amplifier performance of 4.6.5 shall be as follows:

Input Signal	Output Voltage RMS	Phase Reference J200-C to J-200-D
0 (Pins J200-G and J200-H shorted)	1.5 maximum	---
Per 4.3.3.2	13 minimum	Within ±30° of room temperature phase angle
Per 4.3.3.3	15 minimum	---

4.6.8 High temperature.- Each sample unit shall be subjected to a temperature of 71° ±2°C with normal power applied for a period of 4 hours. Power shall be removed for a period of 1/2 hour, maximum. The unit shall meet the requirements of 4.6.3 through 4.6.6 within the following limits:

a. The delay time of 4.6.4.1 shall be from 45 to 85 seconds

b. Leveling amplifier performance of 4.6.5 shall be as follows:

Input Signal	Output Voltage RMS	Phase Reference J200-C to J200-D
0 (Pins J200-G and J200-H shorted)	1.5 maximum	---
Per 4.3.3.2	11.0 minimum	90° ±15° Lead
Per 4.3.3.3	15.0 minimum	---

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4.6.9 Vibration Error.- The unit shall be mounted on a vibration table. The mounting shall simulate service installation orientation, including all holding devices and electrical cables. A complex wave vibration test shall be conducted to determine the ability of the equipment to function in a vibratory environment. During the vibration test, the input vibration to the equipment shall have a power spectral density of 0.015 g^2/cps for each frequency from 20 to 2,000 cps. To accomplish this representation, the loading effect of the unit under test shall be eliminated by proper load compensation. This random vibration shall be applied to the equipment along the three mutually perpendicular axes of the component. The signal source shall be either a prepared magnetic tape or a noise generator. The unit shall be operating throughout the test and shall meet the requirements of the individual tests. The duration of the test shall be 10 - 15 minutes in each axis.

4.6.10 Voltage and frequency variation.- The unit shall be subjected to a voltage of 125V ac and 30V dc at 380 cps and at 420 cps. The unit shall then be subjected to a voltage of 105V ac and 25V dc at 380 cps and at 420 cps. The unit shall be tested in accordance with 4.6.3, 4.6.4, and 4.6.5 in all four cases. The tests shall be conducted while the unit is at the voltage and frequency extremes.

4.6.11 Dielectric strength.- Each isolated circuit shall have a potential of 500V dc applied between it and all other such circuits and the case for a period of 10 seconds. They shall exhibit a resistance of not less than 50 megohms. The applied voltage shall have a source impedance of 1 megohm, shunted by 50 UUF maximum. When applying the dielectric test, pins F, G, H, S, T and U shall be shorted together.

4.6.12 Radio noise interference.- The radio noise interference test shall be conducted in accordance with MIL-I-6181.

4.6.13 Altitude-Temperature.- The unit shall be subjected to a high altitude-low temperature test in accordance with method 504 of MIL-STD-810 for class 2, except step 5 shall be conducted at 85°C for 4 hours, step 7 shall be conducted at 85°C, and steps 11 and 12 shall be at 80,000 ft. The unit shall then meet the individual tests.

4.6.14 Voltage-temperature.- For the purpose of these tests, the following conditions shall apply.

a.	High Voltage	117.5 ±0.5V rms 400 cps 28.5 ±0.5V dc

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b. Low Voltage 108.5 ±0.5V rms 400 cps 25.0 ±0.5V dc

c. Temperature -54° to +72°C

4.6.14.1 Low Voltage - Low temperature.- The unit shall be tested in accordance with 4.6.3 through 4.6.6 at the voltage specified in 4.6.14.b and at -54° C, within the following limits:

a. Voltage outputs of 4.6.3 shall be as follows:

J200 Pins	Output Voltage	Phase
c to a	From 97 to 103 percent of input voltage	180° ±5° (Reference J200-A to J200-D)
P to a	From 20.5 to 26.5 percent of input voltage	0° ±5° (Reference J200-C to J200-D)

b. The delay time of 4.6.4.1 shall be from 45 to 100 seconds

c. Voltage output of 4.6.4.1 shall be from 97 to 103 percent of input voltage

d. Leveling amplifier performance of 4.6.5 shall be as follows:

Input Signal	Output Voltage	Phase Reference J200-C to J200-D)
0 (Pins J200-G and J200-H shorted)	1.5 maximum	--
Per 4.3.3.2	13.0 minimum	90° ±15° Lead
Per 4.3.3.3	15.0 minimum	--

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4.6.14.2 High voltage - High temperature.- The unit shall be tested in accordance with 4.6.15.a at +72°C, 4.6.3 through 4.6.6 at the voltage specified in 4.6.14.a and at 71°C within the following limits:

a. Voltage outputs of 4.6.3 shall be as follows:

J200 Pins	Output Voltage	Phase
c to a	From 97 to 103 percent of input voltage	180° ±5° (Reference J200-A to J200-D)
P to a	From 20.5 to 26.5 percent of input voltage	0° ±5° (Reference J200-C J200-D)

b. The delay time of 4.6.4.1 shall be from 45 to 85 seconds

c. Voltage output of 4.6.4.1 shall be from 97 to 103 percent of input voltage

d. Leveling amplifier performance of 4.6.5 shall be as follows:

Input Signal	Output Voltage	Phase (Reference J200-C to J200-D)
0 (pins J200-G and J200-H shorted	1.5 maximum	--
Per 4.3.3.2	11.0 minimum	90° ±15° Lead
Per 4.3.3.3	15.0 minimum	--

4.6.15 Acoustical noise.- The unit shall be subjected to an acoustical noise test in accordance with MIL-STD-810, method 515, procedure I, grade A.

4.6.16 Dynamic shock.- The dynamic shock test shall be conducted in accordance with method 516, procedures I and IV of MIL-STD-810. Following procedure I, the units shall be subjected to the individual tests.

4.6.17 Vibration

4.6.17.1 Vibration failure.- The vibration failure test shall be conducted in accordance with method 514 of MIL-STD-810, as modified by figure 5, with the power applied. At the end of this test, the units shall be subjected to the individual tests.

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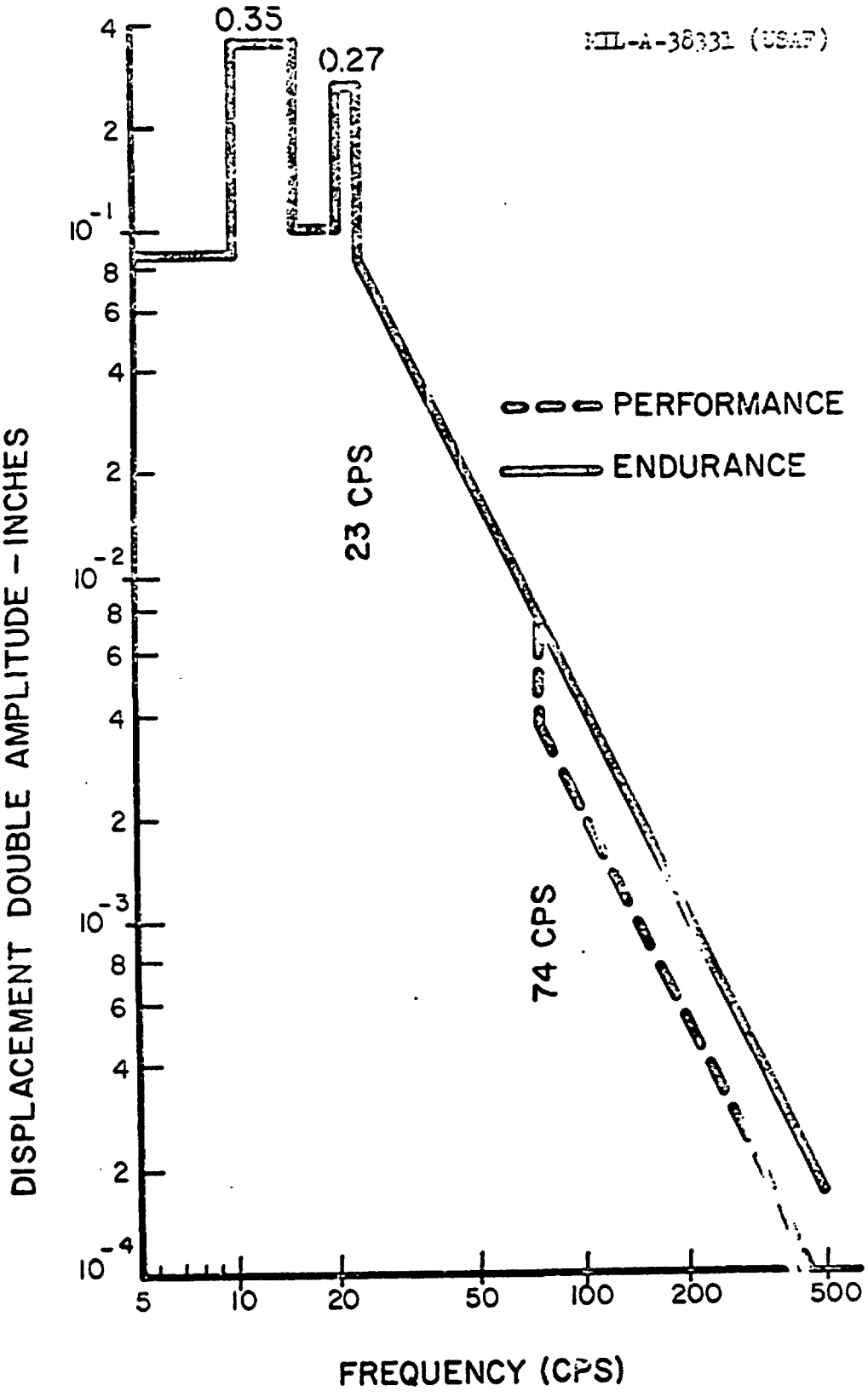


FIGURE 5. VIBRATION TEST CURVES

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4.6.17.2 Vibration endurance.- The vibration endurance test shall be conducted in accordance with method 514 of MIL-STD-810 as modified by figure 5. The frequency shall vary at a logarithmic rate from 5 to 500 cycles/sec and back to 5 cycles in approximately 15 minutes in each of the 3 axes. The equipment shall be operating during this test but need not give specified performance. At the conclusion of the test, the units shall be subjected to the individual tests.

4.6.18 Acceleration.- The acceleration test shall be conducted in accordance with method 513 of MIL-STD-810. The operational test shall be conducted with a test level of 4.5g's in the vertical-up direction. At the conclusion of the operational test, the units shall be subjected to the individual tests. Dummy loads may be used when conducting the structural test.

4.6.19 Environmental tests.- The amplifier-power supply shall be subjected to the following environmental tests in accordance with the specified procedures of MIL-STD-810. At the completion of each test, the unit shall be inspected for damage ~~that would~~ impair operation.

4.6.19.1 Humidity.- The humidity test shall be conducted in accordance with procedure I, method 507.

4.6.19.2 Fungus.- The fungus test shall be conducted in accordance with procedure I, method 508.

4.6.19.3 Salt fog.- The salt fog test shall be conducted in accordance with procedure I, method 509.

4.6.19.4 Sand and dust.- The sand and dust test shall be conducted in accordance with procedure I, method 510.

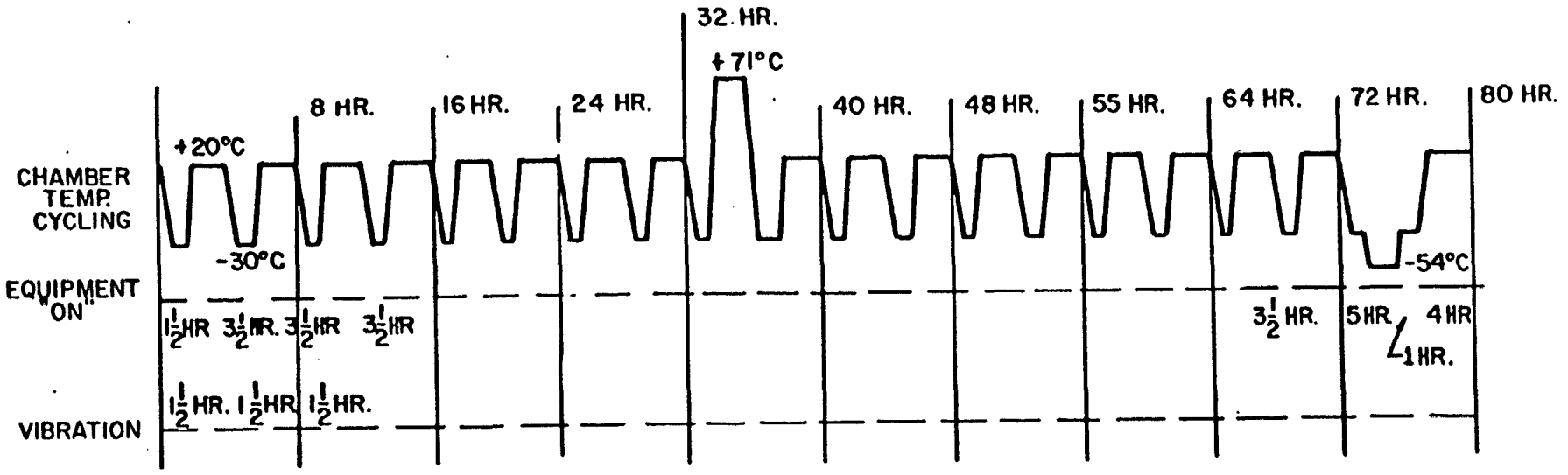
4.6.19.5 Rain.- The rain test shall be conducted in accordance with method 506, procedure I.

4.6.20 Reliability.- The amplifier-power supply when properly connected and power applied as shown on figure 1 shall be subjected to the reliability environmental cycles shown on figure 6. After each 160 hours of operation, the unit shall meet the tests specified in 4.6.1 through 4.6.6.

4.6.20.1 Longevity.- Upon completion of the test specified in 4.6.20, 3 of the leveling amplifiers shall continue to be tested through a longevity test for a total of 2,000 hours on each unit.

4.6.20.2 Relevancy criteria.- All failures shall be classified as relevant failures unless approved for exclusion by the procuring activity.

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a. ENVIRONMENTAL CYCLE REPEATS EVERY 80 HRS.

b. TEMP. RATE OF CHANGE OF CHAMBER SHALL BE 0.1°C - 0.5°C PER SECOND.

c. VIBRATION LEVEL 0.2G. AT 20 TO 30 CPS.

d. "OFF" PERIODS FOR EQUIPMENT "ON" & "OFF" ARE $\frac{1}{2}$ HR. (EXCEPT 76 TH. HOUR).

e. MONITORING OF SYSTEM OUTPUTS FOR CATASTROPHIC-TYPE FAILURES PERFORMED DURING EACH "ON" CYCLE.

f. "SATISFACTORY PERFORMANCE" TESTS ON SYSTEMS & INDIVIDUAL UNITS CONDUCTED EVERY 160 HOURS OF ENVIRONMENT CYCLE TO DETECT PERFORMANCE DEVIATION TYPE FAILURES.

FIGURE 6. RELIABILITY TEST ENVIRONMENT CYCLES

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4.6.20.3 Test information.- Sixty days prior to starting this test, the contractor shall submit detailed test information to the procuring activity for approval. This information shall include but not necessarily be limited to:

a. Equipment used in monitoring performance

b. Periodicity of calibration checks to be made on monitoring equipment

c. Description of environmental test facilities to be used, including recording facilities, method of control of heating and cooling facilities, and vibration equipment

d. Frequency of monitoring.

4.6.20.4 Test records.- Throughout this test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for reproduction. As a minimum, the records shall include the following in accordance with MIL-R-26667: An operating sheet, log of failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the data shall be submitted to the procuring activity within 30 days after the completion of each test phase.

4.6.20.5 Repair of failed units.- In the event of a failure during testing, the failed unit shall be repaired and the test continued.

4.6.21 Maintainability verification.- Maintainability verification shall be in accordance with MIL-M-26512, Appendix A, paragraph entitled "Maintenance Task Test and Demonstration." All tasks selected shall be effectively simulated.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use.- The AM-4080/ASN-55 amplifier-power supply covered by this specification is intended for use in the RF-4C aircraft and is a component of the AN/ASN-55 attitude heading reference set.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted

c. When sampling plan C tests will not be conducted

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d. Levels of packaging and packing desired

e. All failures occurring during testing to be reported to the procuring activity within 2 working days.

6.3 Definition

6.3.1 Failure.- A failure is considered as any electrical or mechanical condition that prevents compliance with the requirements specified herein.

Custodian:
Air Force - 11

Preparing Activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A ☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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